Sneha Patel

Computer Vision | Agentic AI | Machine Learning | IoT Systems | Final-Year B.Tech CSE

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ABOUT ME

I'm a final-year CSE student at Navrachana University building AI systems that work in the real world. I like combining multiple technologies in one project, whether that's a deepfake detector that reached 97.5% accuracy by blending vision transformers with classical image-forensics techniques, an agentic AI platform that plans and retrieves its own context, or a smart-bin system that pairs computer vision with IoT hardware to physically sort waste. I've completed ML and data-analytics internships, led a team to build a transport-operations platform at a hackathon, and hold certifications from AWS, GCP, IBM and Neo4j. I enjoy learning by actually building things.

PROJECTS

DeepShield AI- Deepfake Detection *: PyTorch, ViT, FastAPI, Next.js, OpenCV (97.5% accuracy on test data)*

- Built a deepfake detection system combining four methods, Vision Transformer analysis, Error Level Analysis, DCT frequency analysis and HSV colour statistics, so detection doesn't rely on a single signal.
- The combined model reached 97.5% accuracy on unseen test images, outperforming any single method alone.
- Added Grad-CAM heatmaps for interpretability and FGSM-based perturbations to test and improve robustness against adversarial images.
- Shipped as a full web app: a FastAPI backend for inference and a Next.js frontend with drag-and-drop image upload.

NeuraBin- AI-Powered Smart Waste Intelligence System *: YOLO, Computer Vision, Arduino/ESP, IoT Sensors*

- Built an AI + IoT waste-segregation system that combines computer vision, sensor-based monitoring and automated physical sorting.
- Trained detection across five categories, biodegradable, recyclable, hazardous, non-biodegradable and mixed waste.
- Integrated ultrasonic, IR and environmental sensors with servo-controlled mechanisms for automated sorting and bin fill-level monitoring.
- Built as a team project for an AI hackathon focused on deploying computer vision in a real hardware environment; also presented at the SustainX 2026 International Conference.

Arcanum AI- Agentic AI Platform *: Python, LangGraph, LLMs, RAG, Vector Databases*

- Designed a modular agentic AI workflow for task decomposition, reasoning and contextual information retrieval.
- Implemented retrieval-augmented generation to ground agent responses in domain-specific knowledge.
- Structured the system around specialised agents and workflow orchestration for extensibility and reliable task execution.

TransitOps, Transport Operations Management Platform *: React, Node.js, PostgreSQL, Prisma, REST APIs*

- Led development, as Team Lead, of a centralized platform for managing transport operations and lifecycle workflows at the Odoo Hackathon 2026.
- Contributed across frontend, backend and database layers to bring operational workflows into one unified system.
- Designed, built and integrated the platform end-to-end under hackathon time constraints.

Local Vegetable Variety Classifier *: PyTorch, ResNet18, Custom CNNs, Flask (6 CNN architectures compared)*

- Compared six CNN designs, DeepCNN, DepthwiseCNN, GAP-CNN, Multiscale, Residual and a fine-tuned ResNet18, to find the best fit for fine-grained image classification.
- Wrote a preprocessing pipeline for background removal, class balancing and augmentation to handle noisy, real-world lighting.
- Packaged the best-performing model into a Flask web app for photo-based variety prediction.

AI Supplier Ranking Agent *: Python, Regression Models, Ranking Algorithms*

- Built a tool that scores and ranks suppliers automatically using regression-based scoring and a verification algorithm.
- Presented at a hackathon, where the CEO of Mesh Works praised the approach and its practical thinking around supplier evaluation.

Other projects: Real-Time Health Monitoring System (Fitbit API + Scikit-learn) · Financial Sentiment Analysis (TF-IDF + Logistic Regression) · LeadShift Lead-Generation Platform · BakeMyDay Online Bakery System (Java, JDBC, MySQL, JSP) · IoT automation builds (Arduino, PIR/HC-SR04 sensors, servo motors).

EXPERIENCE

Data Specialist : Internship, CSRBOX

Jan 2024 – Mar 2024

- Built an ML pipeline for behavioural and sentiment-based risk prediction using Python, Scikit-learn and NLP.
- Applied Neo4j graph analysis to surface relationships and patterns not readily visible in tabular data.
- Documented findings and presented model results through technical reports for the team.

Machine Learning Intern : Brainy Beam Technologies, Ahmedabad

Jun 2023 – Aug 2023

- Built a vehicle recommendation system that suggests cars based on customer preferences and past behaviour.
- Applied predictive analytics and feature engineering, and wrote progress reports for management.

EDUCATION

B.Tech - Computer Science Engineering - Navrachana University, Vadodara

Jul 2024 – May 2027 · CGPA: 8.57

Diploma - Computer Engineering - Government Polytechnic for Girls, Ahmedabad

Sep 2021 – 2024 · CGPA: 9.42 · Final-semester SPI: 10.0

SKILLS

AI / Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, CNNs (ResNet, MobileNet, custom), Vision Transformers, YOLO, FGSM, Grad-CAM

Generative & Agentic AI: LangChain, LangGraph, LLMs/SLMs, RAG, vector databases, prompt engineering, multi-agent workflow orchestration

Computer Vision & NLP: OpenCV, image preprocessing, image classification, object detection, Error Level Analysis, DCT analysis, TF-IDF, sentiment analysis, anomaly detection, time-series

Backend / Web: FastAPI, Flask, React, Next.js, Node.js, TypeScript, REST APIs, Java Servlets, JSP, JDBC, HTML, CSS

Data / Databases: Pandas, NumPy, SQL, PostgreSQL, MySQL, MongoDB, Neo4j, Cypher, Matplotlib, Seaborn

IoT / Systems: Arduino, ESP, Raspberry Pi, HC-SR04, IR and environmental sensors, PIR sensors, servo motors, embedded systems, Fitbit API

Programming Languages: Python, Java, C++, C, JavaScript

CERTIFICATIONS

Cloud & AI: AWS Academy – Machine Learning Foundations · Google Cloud – Data Engineer Pro · Generative AI Essentials: Using LLMs to Work with Data

Data Science: IBM – Data Analysis Using Python · IBM Data Analytics Internship Program (2024–25) · Natural Language Processing and Computer Vision

Databases & Emerging Tech: Neo4j Certified Professional · Explore Emerging Tech – IBM SkillsBuild

IoT, Security & Management: Cisco – Introduction to IoT and Digital Transformation · Cisco – Introduction to Cybersecurity · IBM SkillsBuild – Project Management Fundamentals · Programming Foundations: Beyond the Fundamentals

ACHIEVEMENTS & LEADERSHIP

- Team Lead, Odoo Hackathon 2026 - led development of the TransitOps platform.
- IndiGenius AI Hackathon - developed the NeuraBin AI + IoT waste-segregation solution.
- SustainX 2026 International Conference - contributed to research on AI-enabled smart waste management.
- WSRO Nationals - Robot Race competitor, demonstrating programming, hardware integration and teamwork.

LANGUAGES

English (Professional) · Hindi (Fluent) · Gujarati (Native) · German